

4.3 Rates of Change Other Than Motion

CA #1

Calculus

Name: _____

1. The tide removes sand from a beach at a rate modeled by the function R , given by

$$R(t) = 1 + 6 \sin\left(\frac{2\pi t}{25}\right).$$

A pumping station adds sand to the beach at a rate modeled by the function S , given by $S(t) = \frac{18t}{1+4t}$. Both $R(t)$ and $S(t)$ have units of cubic yards per hour and t is measured in hours for $0 \leq t \leq 10$. Find the rate at which the total amount of sand on the beach is changing at time $t = 6$.

2. Tourists visiting an island resort contracted a mystery illness over a 45-day period. The health authorities recorded the rate of new cases per day and some of the rates are listed in the table below.

t Day	$N(t)$ New cases per day
2	3
6	8
10	15
15	30
25	100
35	50
40	22
45	10

- a. Use the table to estimate $N'(20)$. Show the computations that lead to your answer. Indicate units of measure.
- b. After studying the spread of the disease, the health department authorities decided they could model the number of new cases per day with the model $R(t) = \frac{80000e^{-0.2t}}{(1+200e^{-0.2t})^2}$ for $0 \leq t \leq 50$ days. The disease is considered eradicated when the number of new cases per day does not exceed 5. Use $R(t)$ to find on what day this will occur.
- c. Find $R'(20)$. Using appropriate units, explain the meaning of your answer in terms of the decay rate of the substance.
3. The number of gallons, $P(t)$, of a pollutant in a lake changes at the rate $P'(t) = 2 - 4e^{-0.3\sqrt{t}}$ gallons per day, where t is measured in days. There are 100 gallons of pollutant in the lake at time $t = 0$. Is the amount of pollutant increasing at time $t = 7$? Why or why not?

4.

t (minutes)	$R(t)$ (gallons per minute)
0	15
20	25
30	45
50	60
60	70
90	75

Use the data from the table to find an approximation for $R'(40)$. Show the computations that lead to your answer. Indicate units of measure.

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5. A store is having a 10-hour sale. The total number of shoppers who have entered the store t hours after the sale begins is modeled by the function E defined by $E(t) = 0.2t^4 - 10t^3 + 50t^2$ for $0 \leq t \leq 10$. At what rate are shoppers entering the store 4 hours after the start of the sale?

Answers to 4.3 CA #1

1. $S(6) - R(6) = -2.668 \text{ yd}^3 \text{ per hour}$	2a. $\frac{N(25)-N(15)}{25-15} = 7 \text{ new cases per day per day}$
2b: $R(t) = 5$ when $t \approx 48.274$. According to the model, the disease will be eradicated on the 49 th day.	2c: On the 20 th day, the rate of new cases per day is increasing by 7.6966 cases per day per day.
	3. $P'(7) \approx 0.191 > 0$ so the amount of pollutant is increasing at this time.
4. $\frac{R(50)-R(30)}{50-30} = 0.75 \text{ gallons per min}^2$	5. $E'(4) = -28.8 \text{ shoppers per hour}$